

2. Cystic Fibrosis (CF) is a serious medical condition which affects the lungs. It results from a mutation in the DNA of a single gene which gives rise to a recessive allele.

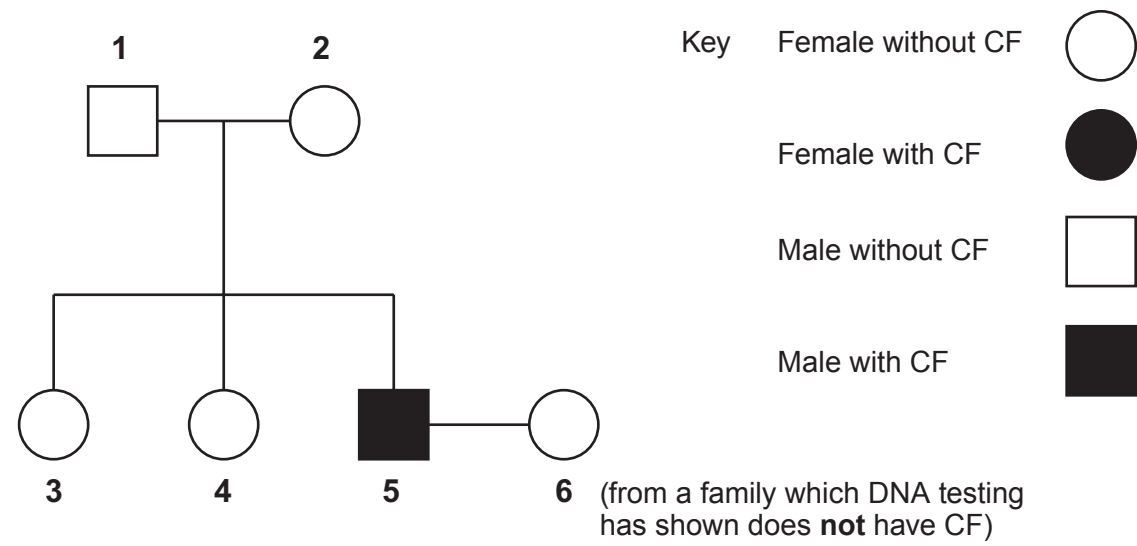
(a) Explain the meaning of the term *recessive allele*. [2]

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(b) The diagram below shows part of the family tree of a family which has one member who has CF.



(i) State the effect of a mutation on DNA. [1]

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(ii) From the family tree:

I. State the numbers of **two** individuals who are known to be heterozygous for the allele which causes CF. [1]

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II. Suggest what advice a genetic counsellor would give individuals **5** and **6** about the chance of them having a child with CF. Explain your answer. [2]

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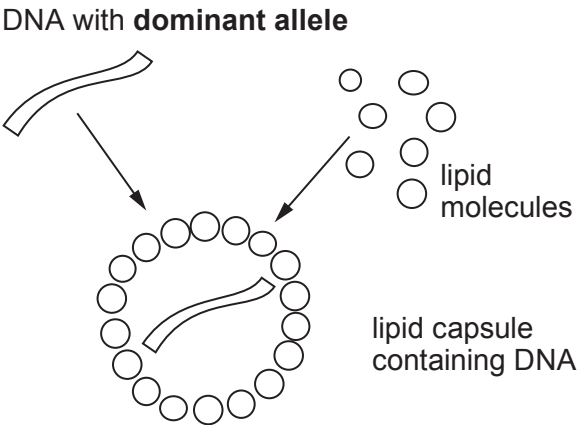
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- (c) CF is usually treated by intensive physiotherapy which can be very stressful. Medical scientists are now working to develop an alternative treatment called *gene therapy*. This process is summarised in the flow chart below.



Physiotherapy for CF



Gene therapy for CF

- (i) Lipid capsules are introduced into the lungs. State how this would be done. [1]

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- (ii) In order for this therapy to be successful two further difficulties must be overcome. Describe these **two** difficulties and suggest why this therapy is **not** a permanent cure. [3]

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